

Hedgewise

Risk Minimized Portfolio Construction through Algorithmic Modeling

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INTRODUCTION

I started investing at age 12 when my mother, a 25-year market veteran, introduced me to stocks. This early experience shaped my understanding of long-term wealth creation and revealed a troubling reality:

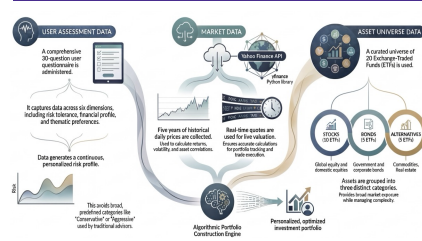
•**False Diversification:** S&P 500 now dangerously concentrated — Magnificent Seven represents ~35% of market cap

•**Knowledge Gap:** Retail investors lack access to institutional-grade tools and educational resources

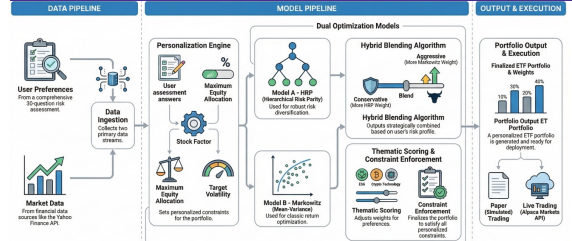
This project was born from a simple question: why should sophisticated portfolio optimization be reserved for Wall Street?

<https://github.com/pratik456/Risk-Minimized-Portfolio-Construction-through-Algorithmic-Modeling>
<https://portfolio-frontend-57326055677.us-central1.run.app/>

DATA



PIPELINE



Market: 5 years of daily prices from yfinance for 20 ETFs (10 stocks, 5 bonds, 5 alternatives)

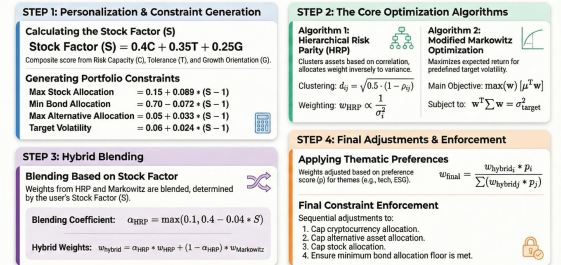
User: 30-question risk assessment capturing 6 dimensions (risk tolerance, financial profile, thematic preferences)

Risk Profiling: 30-question assessment → single risk score determining asset allocation and optimization priority

Hybrid Algorithm: Blends Hierarchical Risk Parity (stability) + Markowitz Optimization (returns) based on user's risk profile

Execution: Paper or live trading via Alpaca Markets API

MODEL



HRP Algorithm: Clusters correlated assets, weights inversely to risk — stable during market stress

Markowitz Algorithm: Maximizes returns for target volatility — optimizes risk-adjusted performance

Personalization: Blends algorithms by risk score (conservative → HRP, aggressive → Markowitz), applies thematic preferences (tech, ESG, dividends), enforces asset allocation limits

RESULTS

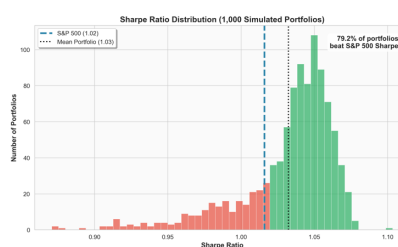
Table 10: Algorithm Performance Comparison (Moderate Risk, 2019-2024)

Metric	HRP Only	Markowitz Only	Hybrid	S&P 500
Annualized Return	2.7%	4.8%	17.1%	18.4%
Annualized Volatility	6.3%	6.5%	17.7%	19.9%
Sharpe Ratio	0.43	0.75	0.97	0.93
Maximum Drawdown	-15.3%	-14.7%	-26.7%	-33.7%



Table 9: Portfolio Outputs for Three User Profiles (All Risk Score = 6.0)

Profile	Stocks	Bonds	Alts	Top ETFs	Return	Vol	Sharpe
Tech-Focused	83%	10%	7%	QQQ, VOO, BITO	10.9%	18.2%	0.49
Income-Focused	46%	42%	12%	SCHD, VTV, BND	6.9%	9.4%	0.54
Global Diversifier	61%	22%	17%	VXUS, VWO, GLD	8.3%	12.1%	0.52



Backtest Performance (2019-2024): Sharpe ratio 0.97 vs S&P 500's 0.93 | Max drawdown 26.5% vs 33.7%

Monte Carlo (1,000 portfolios): 79.2% beat S&P 500 Sharpe ratio | Consistently lower drawdown during market crashes

Risk Reduction: 7.2 percentage point improvement in max drawdown while maintaining competitive returns

CONCLUSION

Hedgewise democratizes institutional-grade portfolio optimization for retail investors with minimum risk, with future work including genetic algorithms, reinforcement learning for dynamic rebalancing, and automated tax-loss harvesting.